Git fundamental

1) Git init?

Is a git command used to initialize a new git repository in a project folder. It allows git to track changes in your file.

Ex:- MKdir myproject
2> create a project folder

cd my project
2> Open the project folder

o/p: Initialized empty git reposito -ry

git init
2> Initialize git in that folder

Remember: Start tracking a project with git.

2) Git clone:

Is a git command used to copy an existing git repository from github.

Ex:- git clone https://github.com/username/repository.git

Remember: copy an existing repository to your computer.

3) Git status:

Is a Git command used to check the current State of your repository

Ex:- Git Status.

Remember: check the current state of your git repository

4) Add

Git command used to add new or modified files to the staging area before committing them.

Ex:- Git add index.html

Remember: prepare files for commit.

5) commit

Git command used to save staged changes to the repository with a descriptive message

Ex:- git commit -m "Added Homepage"

Explanation.

- git commit:- save the staged changes in the repo history.
- -m :- Adds a message describing the changes.
- "Added homepage": Describes what you changed

Remember: Save staget changes with a messa -ge

6) push:-

It is a bjit command used to upload your local commits to a remote repository, such as a git hub.

Ex:- git push origin main

Explanation.

git push - uploads your committed changes.

origin - the name of the remote repo.

main - the branch where you wanto to upload your changes.

Remember: upload your committed code to git hub.

7) pull

It is a bjit command used to download the latest changes from a remote repository, such as git hub, and integrate them into your current local branch.

Ex:- bjit pull origin main.

Explanation.

- git pull:- fetches and integrates remote change
- origin. the name of the remote repo.
- main. the branch from which you want to get changes.

Remember: Get the latest changes from github & integrate them into your local project.

8) Fetch.

Git command used to download the latest update from a remote repository without automati-cally merging them into your current branch.

Eg: git fetch origin

Explanation.

git fetch - Downloads updates from the remote repository.

origin - the name of the remote repository.

fetch	pull
Downloads updates but not merge them into your current branch	Downloads updates & integrates them into your current branch.

9) checkout

Git command used to switch branches or restore file to previous state.

Ex:- Switch to an existing branch git checkout feature

Remember: git checkout switch branches.

NOTE:- Git switch is the modern, clearer command for switching branches git checkout is still widely used

10) Merge.

Git command used to combine changes from one branch into another branch.

Ex: git switch main.
git merge feature

Explanation:

git switch main - switches to the main branch.

git merge features - combines the changes for feature into main.

Remember: Combine changes from one branch to another.

11) Rebase

Git command used to move or replay commits from one branch onto another branch, creating a clean -er, more linear project history.

Ex: git switch feature
git rebase main.

Explanation:-

- git switch feature - switches to the feature branch.
- git rebase main - Replays the feature branch's comm -it's on top of main.

Merge	Rebase
combines branch histo - ries	Replays commits onto another base
May create a merge commit	Usually creates a linear history
Preserves existing commits	Create new commit ID's for replayed commits.

Remember: Replay your commits on top of another branch to create a cleaner history

12) Git log

git log is a Git command used to view the commit history of repository, including commit IDs, authors, dates, and commit message.

Ex: git log

o/p: commit a1b23d
Author: Kuvalesh.
Date: Mon Oct 19 10:00 2026
Initial commit.

Remember: View the history of commits in git.

13) git diff

Git command used to compare changes b/w files, commits, or branches.

Ex:- git diff

ex: o/p - Hello world
+ Hello git

- means a line was removed
+ means a line was added

Remember: compare changes in your code.

14) git stash.

Git command used to temporarily save your uncommitted changes so you can work on something else without committing them.

Eg:- git stash.

Useful commands.

```
git stash          # Save changes temporily
git stash list     # View saved stashes
git stash pop      # Restore changes & remove the stash
git stash apply    # Restore changes but keep the stash
```

Remember: Temporairly save your uncommitted chan-ges.

15) Git reset

Git command used to move the current branch to a previous commit or remove changes from the staging area

Eg:- git reset index.html.

Removes index.html from the staging area but keeps your file changes.

Remember: move the branch back or undo staging

11) Git revert

Git command used to undo the changes introduced by a previous commit by creating a new commit

Ex:- git revert HEAD.

Explanation:-

- git revert - Reverses changes safely by creating a new commit
- HEAD: Refers to the current commit.

git reset	git revert
• moves the branch pointer backward	creates a new commit to undo changes.
• can rewrite commit history	preserves commit history
• Use carefully on shared branches	Generally safer for shared branches.

Remember: Undo a commit without deleting it from history

Git cherry-pick.

git cherry-pick is a git command used to apply a specific commit from one branch to another branch without merging the entire branch.

Ex:- git switch main
git cherry-pick a1b2c3d

Explation:- switch main - switches to the main branch.
. git cherry-pick a1b2c3d - Applies the changes from specified commit.

Remember: select & apply a specific commit from another branch.

git tag

Git command used to mark a specific commit.

Ex: git tg v1.0:

Useful commands.

```
git tag          # List all tags
git tag v1.0 #   Create a tag
git to show v1.0 # View tag details
git push origin v1.0 # push the tag to github.
```

Remember: Mark a specific version of your project